

What is probability?

In this video, you will learn

- The definition of probability.
- Why probability is such an important topic.
- How probabilities are often misinterpreted.

Definition of probability

Probability is a numerical description of how likely an event is to occur or how likely it is that a proposition is true. Probability is a number between 0 and 1, where, roughly speaking, 0 indicates impossibility and 1 indicates certainty.

The sum of all probabilities in a set must sum to 1.

Text copied verbatim from
<https://en.wikipedia.org/wiki/Probability>
Mike's addition.

Examples of correct and incorrect interpretations

Statement "There is a 20% probability of rain today."

~~Interpretation 1 It will rain for 20% of the day $\Rightarrow .2 * 24 = 4.8$ hours.~~

Interpretation 2 There is a 1 in 5 chance that it will rain.

~~Interpretation 3 We can be 20% confident that it will rain today.~~

If we could repeat the exact conditions of today a very large number of times (approaching infinity), it would rain on 20% of those days.

Examples of correct and incorrect interpretations

Statement “There is a 20% probability of rain today.”

~~Interpretation 1 It will rain for 20% of the day $\Rightarrow .2 \times 24 = 4.8$ hours.~~

Interpretation 2 There is a 1 in 5 chance that it will rain.

~~Interpretation 3 We can be 20% confident that it will rain today.~~

Confidence is different from probability: We could be 99% confident of a 20% chance of rain. In this case, 20% is our parameter estimate, and the confidence interval might be [19% 21%].

Examples of probabilities



$$p(\text{heads}) = .5$$



$$p(3) = \frac{1}{6} = 16.7\%$$



$$p(Q\heartsuit) = \frac{1}{52} = 1.92\%$$

Why probability is an important topic

Developed by the Bernoulli family and Bayes as a way to help rich people make money from gambling*.

Weather

Insurance (actuarial science)

Medicine (e.g., modeling infectious disease spread)

Physics, chemistry, biology... and all applied sciences

Deep learning / AI

*According to rumors...

When do we need probability?

We need probability when there is uncertainty about the outcome of an event.

Consider:

1. Speed of light vs. speed of sound
2. Height of men vs. height of women
3. Whether a test for a disease indicates the presence of the disease
4. Whether a potato will fall to the ground if you drop it

Probability vs. proportion

In this video, you will learn

- The distinction between probability and proportion.
- See several examples that illustrate the difference.

Probability vs. proportion

Probability: the likelihood of an event occurring or that a statement is true.

Proportion: a fraction of a whole.

Example: dental hygiene

I spend a total of 5.1 minutes each day brushing my teeth, out of 17 waking hours (1020 minutes).

The proportion of my waking day spent brushing my teeth is $5.1/1020 = .005$ (.5%).

The probability that a randomly selected minute of my day involves teeth-brushing is .005.

The probability that I will brush my teeth during the day is 1.

Example: chocolate

I eat 100g of chocolate every Friday.

The proportion of my week that contains a chocolate-eating day is $1/7 = .143$ (14.3%).

The probability that a randomly selected day involves chocolate-eating is .143.

The probability that I will eat chocolate during the week is 1.

Example: COVID-19

3.4% mortality rate (WHO, 3 March 2020, [worldometers.info](https://www.worldometers.info))

~~If I get infected, I have a 3.4% chance of dying from COVID-19 complications.~~

The probability that a randomly selected infected patient will die from COVID-19 complications is .034.

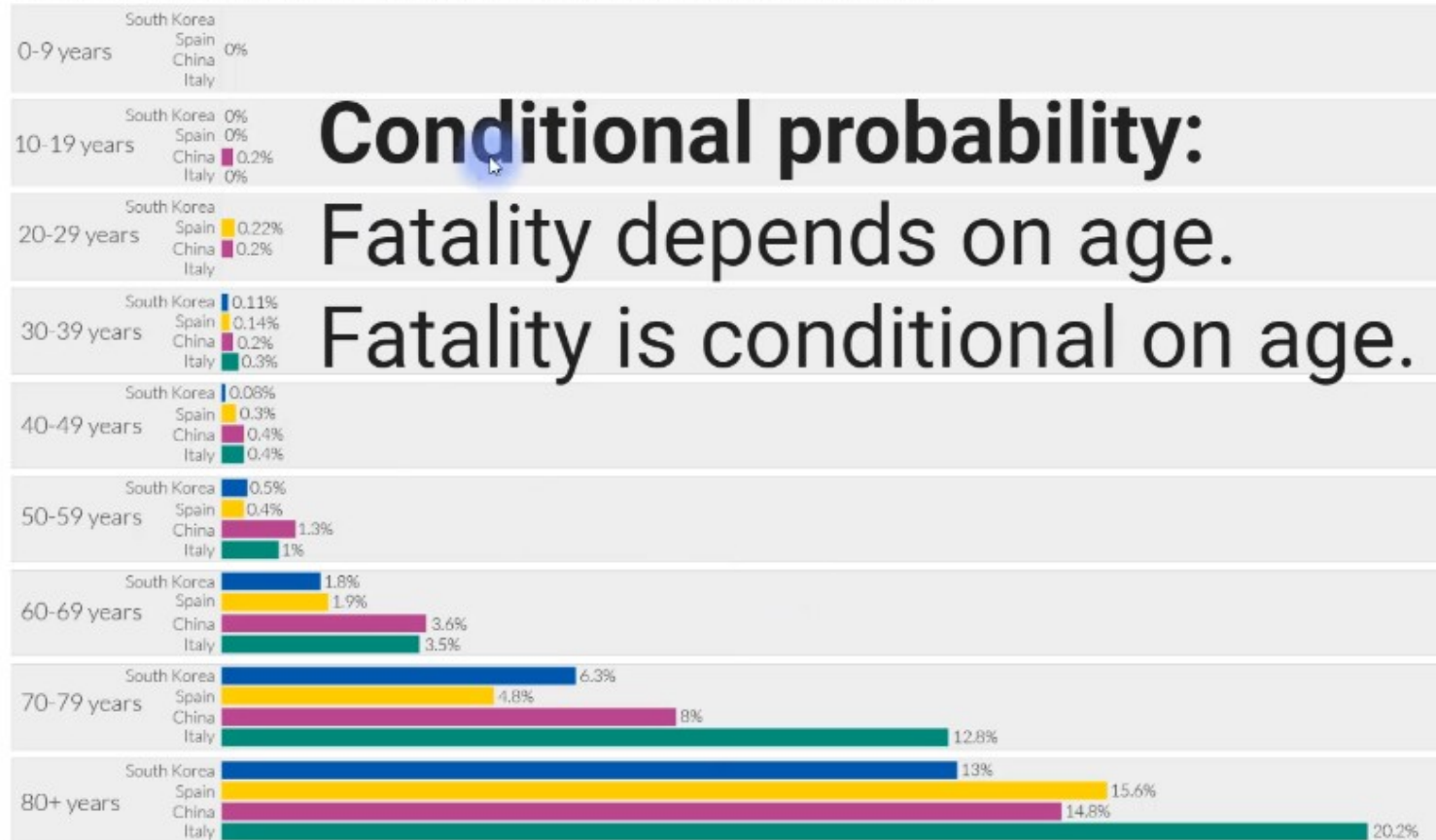
The proportion of infected people who die from complications is .034.

Coronavirus: case fatality rates by age

Case fatality rate (CFR) is calculated by dividing the total number of confirmed deaths due to COVID-19 by the number of *confirmed* cases.

Two of the main limitations to keep in mind when interpreting the CFR:

- (1) many cases within the population are unconfirmed due to a lack of testing.
- (2) some individuals who are infected will eventually die from the disease, but are still alive at time of recording.



Note: Case fatality rates are based on confirmed cases and deaths from COVID-19 as of: 17th February (China); 24th March (Spain); 24th March (South Korea); 17th March (Italy).

Example: coin flips



Proportion of heads = 60%

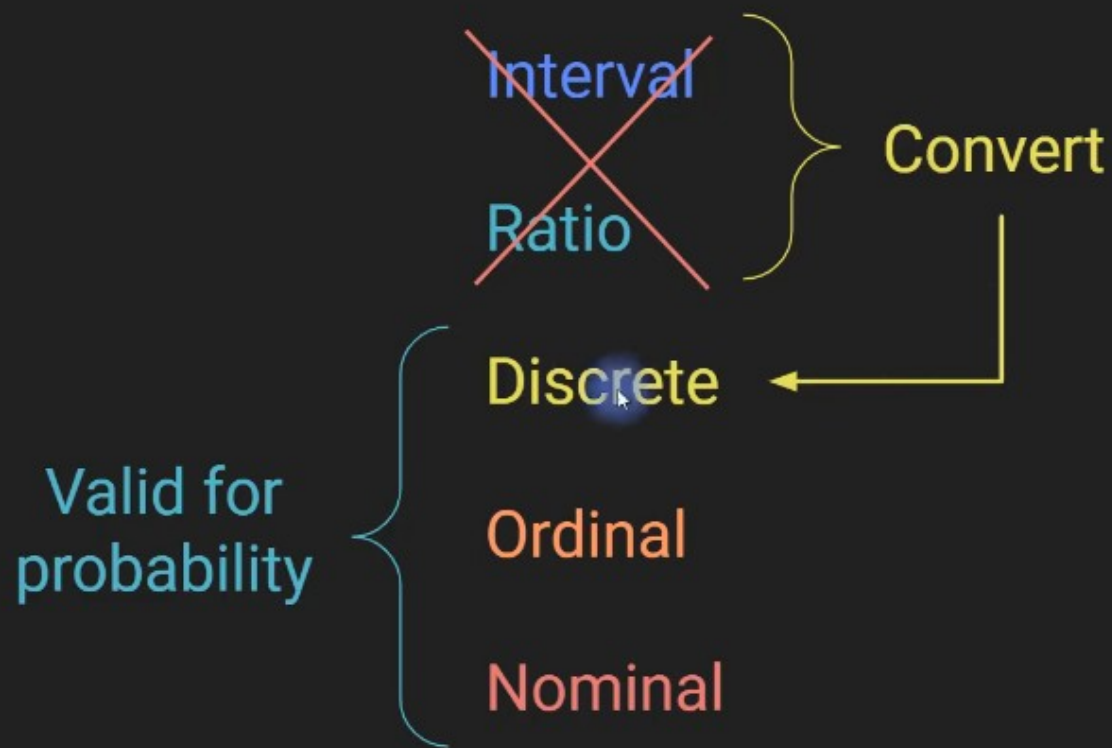
Probability of heads = 50%

Computing probability

In this video, you will learn

- How to make sure data are valid for probability.
- How to convert data into a format from which probabilities can be computed.
- The actual computation of converting occurrences into probabilities .

Data types for probability

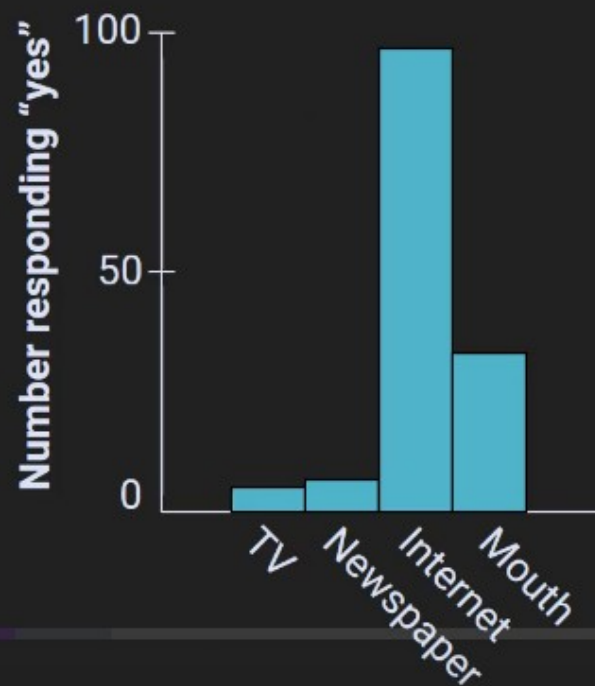


$p(\text{height}=65.2394629384731 \text{ cm})?$

$p(\text{height}>60 \text{ \& } <70 \text{ cm})?$

Data validity for probability

Data must have mutually exclusive labels or bins.
Because total probabilities must sum to 1 (100%).



The actual computation

$$p_i = \frac{100 \times c_i}{\sum c}$$

The **probability** of event ***i*** is the **count of events *i*** divided by the **total number of all events**.

The actual computation



A jar containing:

- 40 blue marbles
- 30 yellow marbles
- 20 orange marbles

What is probability of picking each color at random?

First, determine data validity (data type and exclusivity).
Second, compute probabilities.

The actual computation



A jar containing:

- 40 blue marbles
- 30 yellow marbles
- 20 orange marbles

Data type: nominal
Exclusivity: Yes

Pick blue: $40 \times 100 / 90 = 44.4\%$

Pick yellow: $30 \times 100 / 90 = 33.3\%$

Pick orange: $20 \times 100 / 90 = 22.2\%$

Sanity check:

$44.4 + 33.3 + 22.2 = 99.9\%$

```

import matplotlib.pyplot as plt
import numpy as np

## the basic formula
# counts of the different events
c = np.array([ 1, 2, 4, 3 ])
# convert to probability (%)
prob = 100*c / np.sum(c)
print(prob)

## the example of drawing marbles from a jar
# colored marble counts
blue = 40
yellow = 30
orange = 20
totalMarbs = blue + yellow + orange

# put them all in a jar
jar = np.hstack((1*np.ones(blue), 2*np.ones(yellow), 3*np.ones(orange)))

# now we draw 500 marbles (with replacement)
numDraws = 500
drawColors = np.zeros(numDraws)

for drawi in range(numDraws):
    # generate a random integer to draw
    randmarble = int(np.random.rand()*len(jar))
    # store the color of that marble
    drawColors[drawi] = jar[randmarble]

# now we need to know the proportion of colors drawn
propBlue = sum(drawColors==1) / numDraws
propYell = sum(drawColors==2) / numDraws
propOran = sum(drawColors==3) / numDraws

# plot those against the theoretical probability
plt.bar([1,2,3],[ propBlue, propYell, propOran],label='Proportion')
plt.plot([0.5, 1.5],[blue/totalMarbs, blue/totalMarbs],'b',linewidth=3,label='Probability')
plt.plot([1.5, 2.5],[yellow/totalMarbs,yellow/totalMarbs],'b',linewidth=3)
plt.plot([2.5, 3.5],[orange/totalMarbs,orange/totalMarbs],'b',linewidth=3)

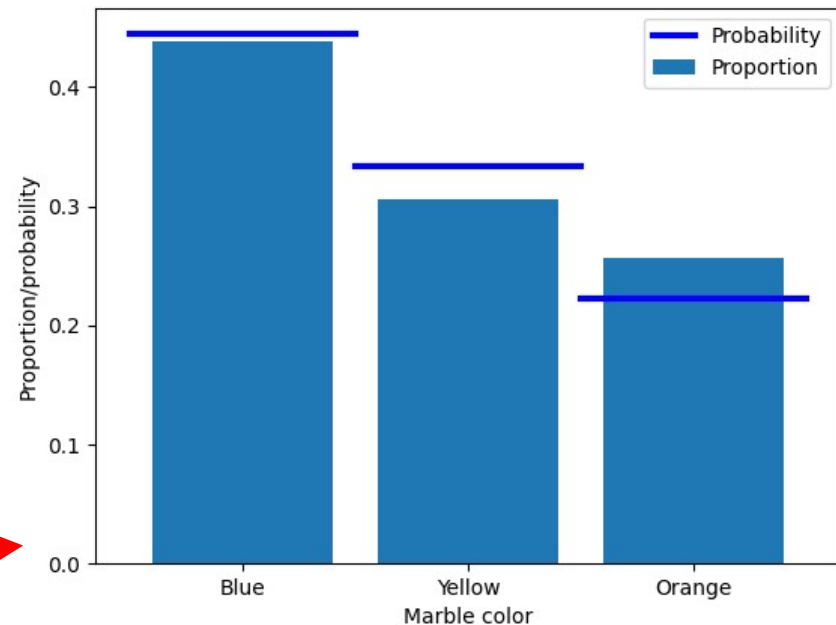
plt.xticks([1,2,3],labels=('Blue','Yellow','Orange'))
plt.xlabel('Marble color')
plt.ylabel('Proportion/probability')
plt.legend()
plt.show()

```

```

Warning: NameError:
[10. 20. 40. 30.]

```



Probability and odds

In this video, you will learn

- The difference between probability and odds.
- How to convert between probability and odds.

Odds

Often discussed in the context of betting or disease.

For example, "The odds are 6:1 against."

$$r = n : m = \frac{n}{m}$$

Odds as a probability ratio

“Odds” is a ratio of the probability of an event not occurring, to the probability of it occurring.

$$r = \frac{1 - p}{p} \Rightarrow p = \frac{1}{1 + r} = \frac{1}{1 + n/m}$$

Odds as a probability ratio

Example: What are the odds of drawing a king in a deck of cards?

$$r = \frac{1 - 4/52}{4/52} = 12 : 1$$

Probability mass vs. density

In this video, you will learn

- The differences among probability, probability mass, and probability density.
- How to interpret and create probability mass and density functions.

Terms

Probability: The chance that an event will occur.

Probability mass: A function that describes probabilities for a set of exclusive *discrete* events.

Probability density: A function that describes probabilities for a set of exclusive *continuous* events.

Probability functions

Unique values of X All values of a variable

$$f(x) = p(X = x)$$

$$f(x_i) = p(X = x_i)$$

$$s.t. \ p(X = x_i) \geq 0$$

$$s.t. \ p(X \neq x_i) = 0$$

$$s.t. \ \sum p(X = x_i) = 1$$

Probability functions

$$f(x) = p(X = x)$$

A function of variable x defined by the probability of different values of x occurring.

$$f(x_i) = p(X = x_i)$$

$$s.t. \ p(X = x_i) \geq 0$$

$$s.t. \ p(X \neq x_i) = 0$$

$$s.t. \ \sum p(X = x_i) = 1$$

Probability functions

$$f(x) = p(X = x)$$

$$f(x_i) = p(X = x_i)$$

The value of the function for each unique element of x is equal to its probability.

$$s.t. p(X = x_i) \geq 0$$

$$s.t. p(X \neq x_i) = 0$$

$$s.t. \sum p(X = x_i) = 1$$

Probability functions

$$f(x) = p(X = x)$$

$$f(x_i) = p(X = x_i)$$

$$s.t. p(X = x_i) \geq 0$$

$$s.t. p(X \neq x_i) = 0$$

Events must be exclusive.

$$s.t. \sum p(X = x_i) = 1$$

Probability functions

$$f(x) = p(X = x)$$

$$f(x_i) = p(X = x_i)$$

$$s.t. \ p(X = x_i) \geq 0$$

$$s.t. \ p(X \neq x_i) = 0$$

$$s.t. \ \sum p(X = x_i) = 1$$

All probabilities must sum to 1.

Probability mass function

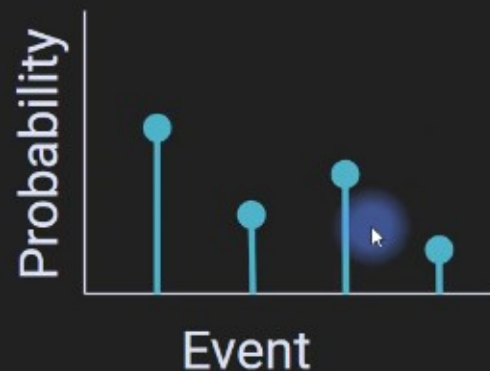
Probability mass: A function that describes probabilities for a set of exclusive *discrete* events.



Lines can be OK
for numeric data.

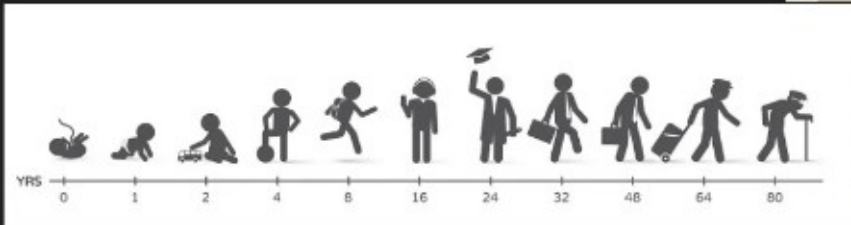
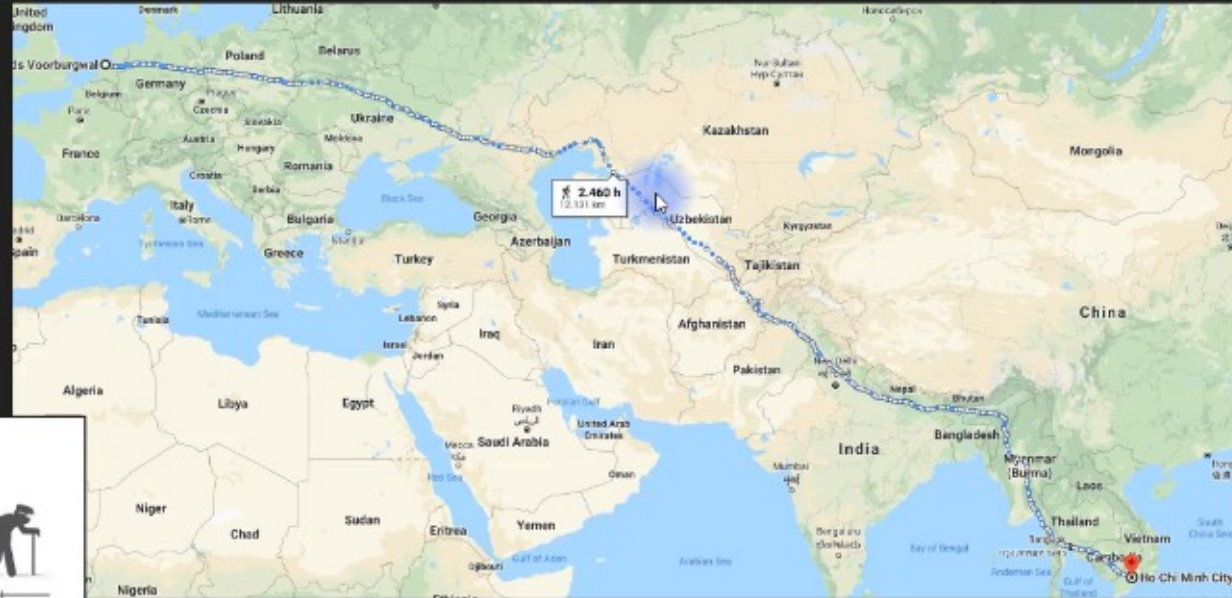
Probability mass function

Probability mass: A function that describes probabilities for a set of exclusive *discrete* events.



Probability density function

Probability density: A function that describes probabilities for a set of *exclusive continuous* events.

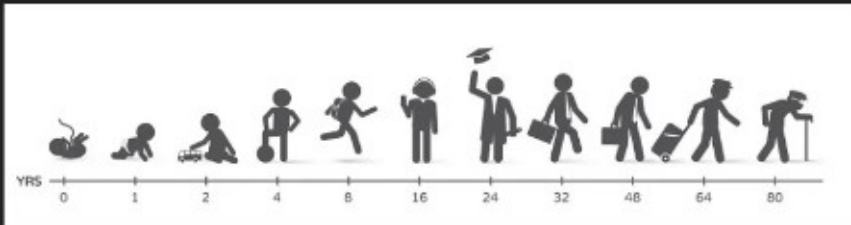


Probability density function

Probability density: A function that describes probabilities for a set of *exclusive continuous* events.

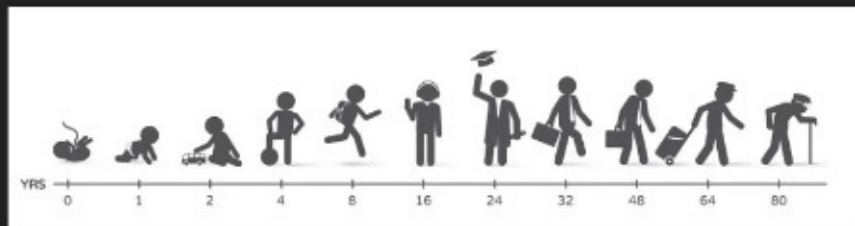


$$P(a \leq X \leq b) = \int_a^b f(X) dx$$



Probability density function

Probability density: A function that describes probabilities for a set of *exclusive continuous* events.



$$P(X \leq b) = \int_{-\infty}^b f(X) dx$$



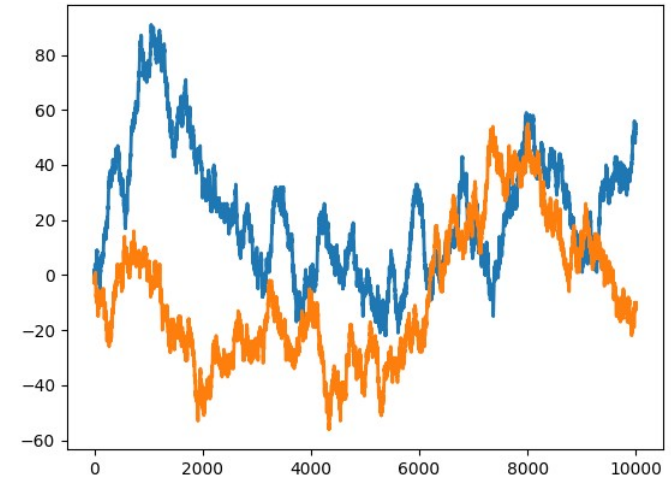
```
import matplotlib.pyplot as plt
import numpy as np

## re-run marble jar code from "compute probabilities"

## compute empirical probability function

# continous signal (technically discrete!)
N = 10004
datats1 = np.cumsum(np.sign(np.random.randn(N)))
datats2 = np.cumsum(np.sign(np.random.randn(N)))

# let's see what they look like
plt.plot(np.arange(N), datats1, linewidth=2)
plt.plot(np.arange(N), datats2, linewidth=2)
plt.show()
```

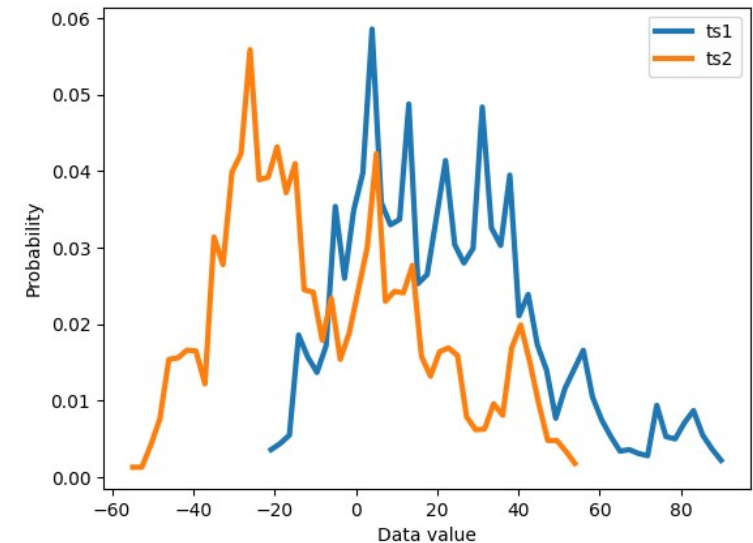


```
# discretize using histograms
nbins = 50

y,x = np.histogram(datats1,nbins)
x1 = (x[1:]+x[:-1])/2
y1 = y/sum(y)

y,x = np.histogram(datats2,nbins)
x2 = (x[1:]+x[:-1])/2
y2 = y/sum(y)

plt.plot(x1,y1, x2,y2,linewidth=3)
plt.legend(('ts1','ts2'))
plt.xlabel('Data value')
plt.ylabel('Probability')
plt.show()
```



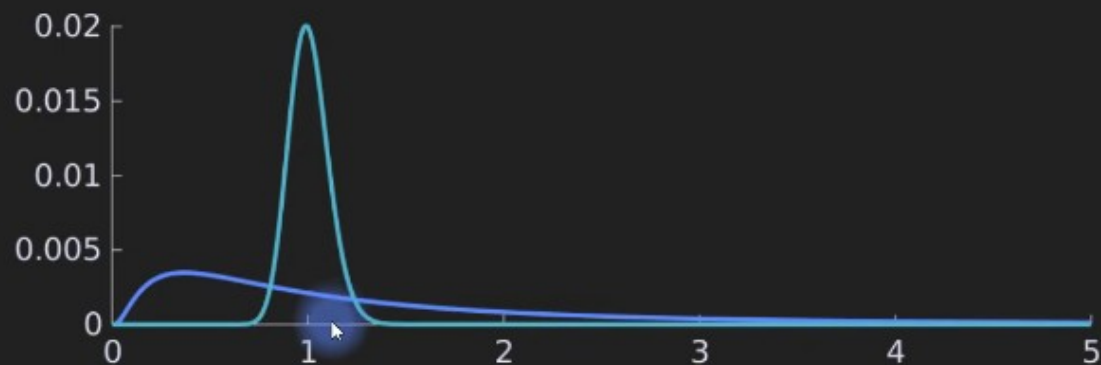
Cumulative distribution functions

In this video, you will learn

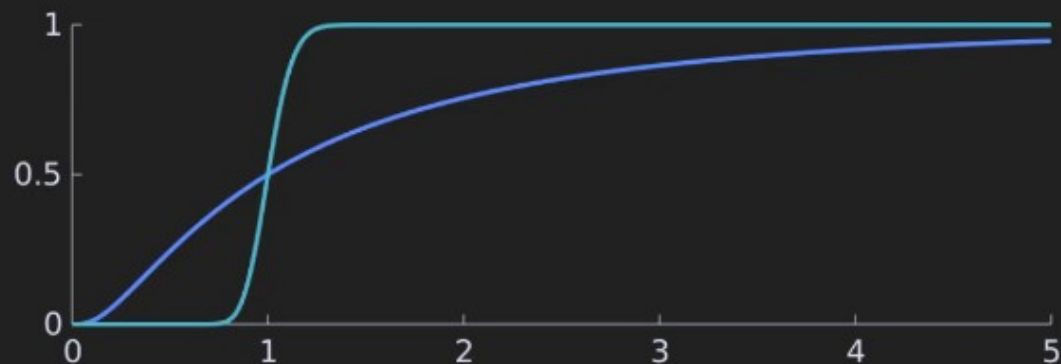
- The difference between a probability density function (or distribution) and a cumulative probability function.
- How to interpret *cdfs*.

Cumulative distribution function (cdf)

Probability
density functions



Cumulative
distribution
functions



Cumulative distribution function (cdf)

A cdf is the cumulative sum (or integral) of the probability distribution (or density).

The y-axis value at each x-value is the sum of all probabilities to the left of that x-value.

A cdf starts at 0 and increases monotonically to 1. The sum of the cdf is more than 1.

$$C(x_a) = p(X \leq x_a)$$

$$C(x_a) = \int_{-\infty}^a p(x_t) dt$$

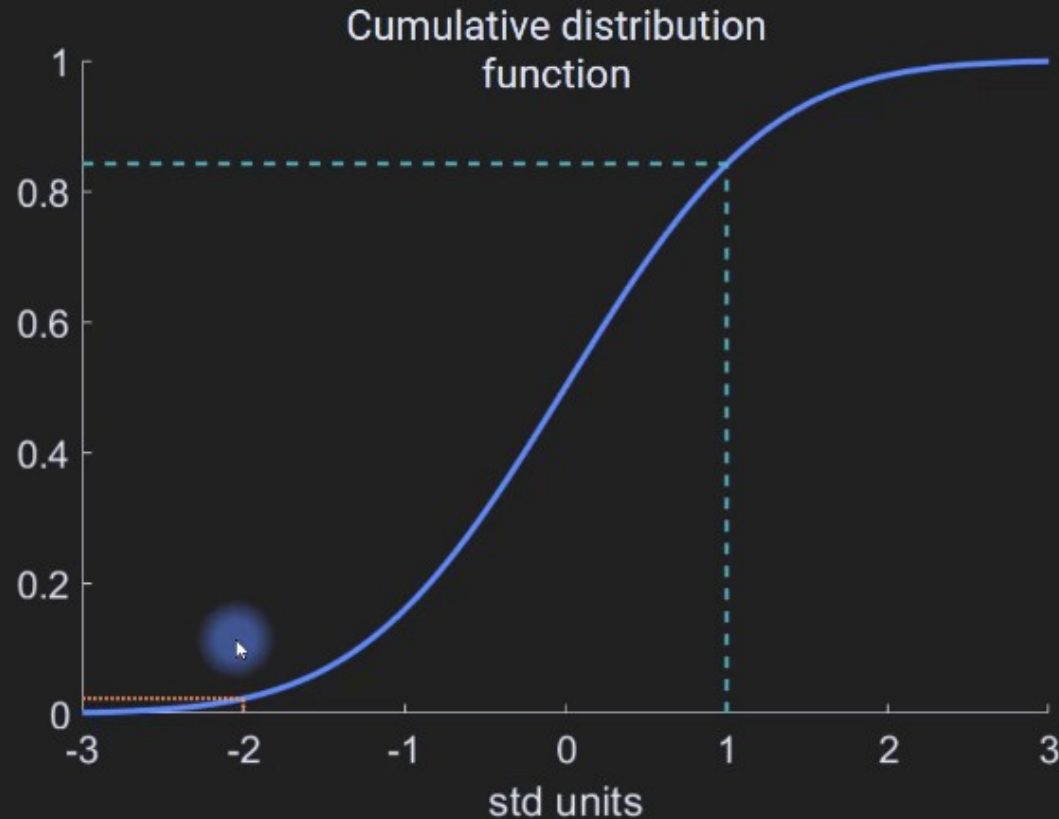
$$C(x_a) = \sum_{i=1}^a p(x_i)$$

Cumulative distribution function (cdf)

cdf's are used to evaluate the probability of obtaining a value up to X or at least X .

Example: *What is the probability of getting at least 1 std higher on the SATs than average?*

Example: *What is the probability of an elephant weighing less than 2 std below the average?*



```

import matplotlib.pyplot as plt
import numpy as np
import scipy.stats as stats

## example using log-normal distribution

# variable to evaluate the functions on
x = np.linspace(0,5,1001)

# note the function call pattern...
p1 = stats.lognorm.pdf(x,1)
c1 = stats.lognorm.cdf(x,1)

p2 = stats.lognorm.pdf(x,.1)
c2 = stats.lognorm.cdf(x,.1)

# draw the pdfs
fig,ax = plt.subplots(2,1,figsize=(4,7))

ax[0].plot(x,p1/sum(p1)) # question: why divide by sum here?
ax[0].plot(x,p1/sum(p1), x,p2/sum(p2))
ax[0].set_ylabel('probability')
ax[0].set_title('pdf(x)')

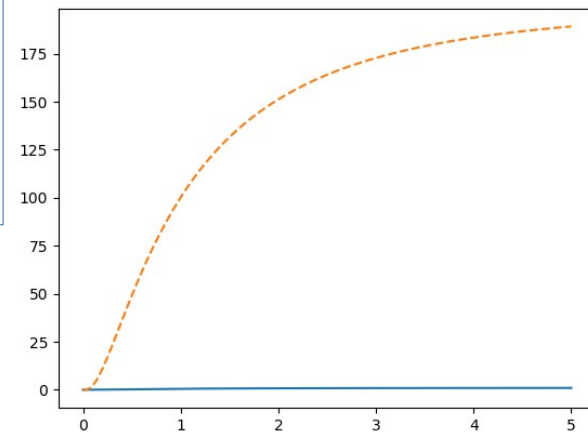
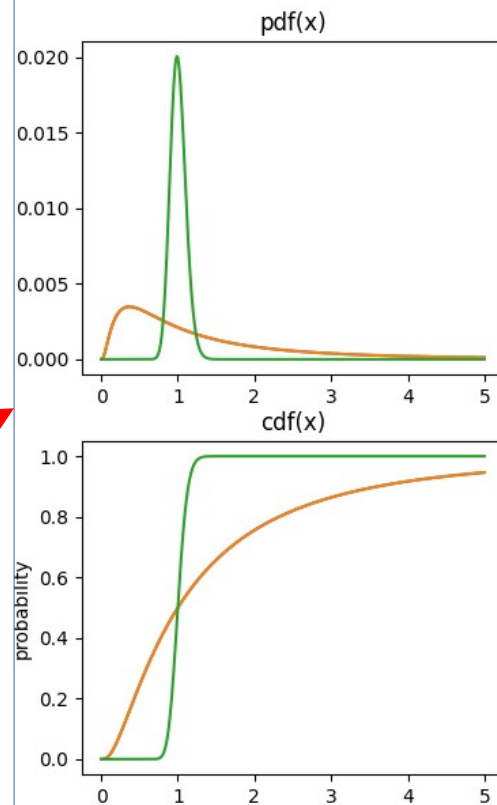
# draw the cdfs
ax[1].plot(x,c1)
ax[1].plot(x,c1, x,c2)
ax[1].set_ylabel('probability')
ax[1].set_title('cdf(x)')
plt.show()

## computing the cdf from the pdf

# compute the cdf
c1x = np.cumsum( p1 )

plt.plot(x,c1)
plt.plot(x,c1x,'--')
plt.show()

```



Creating sample estimate distributions

In this video, you will learn

- Where distributions of sample estimates come from.
- Why random and representative sampling is important.

How to create a sample distribution

Scientific question:

How tall are giraffes?

What is the population parameter of giraffe height?

This question is impossible to answer!

Why?

Because we cannot measure ALL giraffes.

Instead, we measure the heights of a sample of giraffes.

How to create a sample distribution

Scientific question

A **group of giraffes** is **called** a "tower," a **group** of hippos is **called** a "bloat," and a **group** of hyenas is **called** a "cackle." Jan 12, 2012

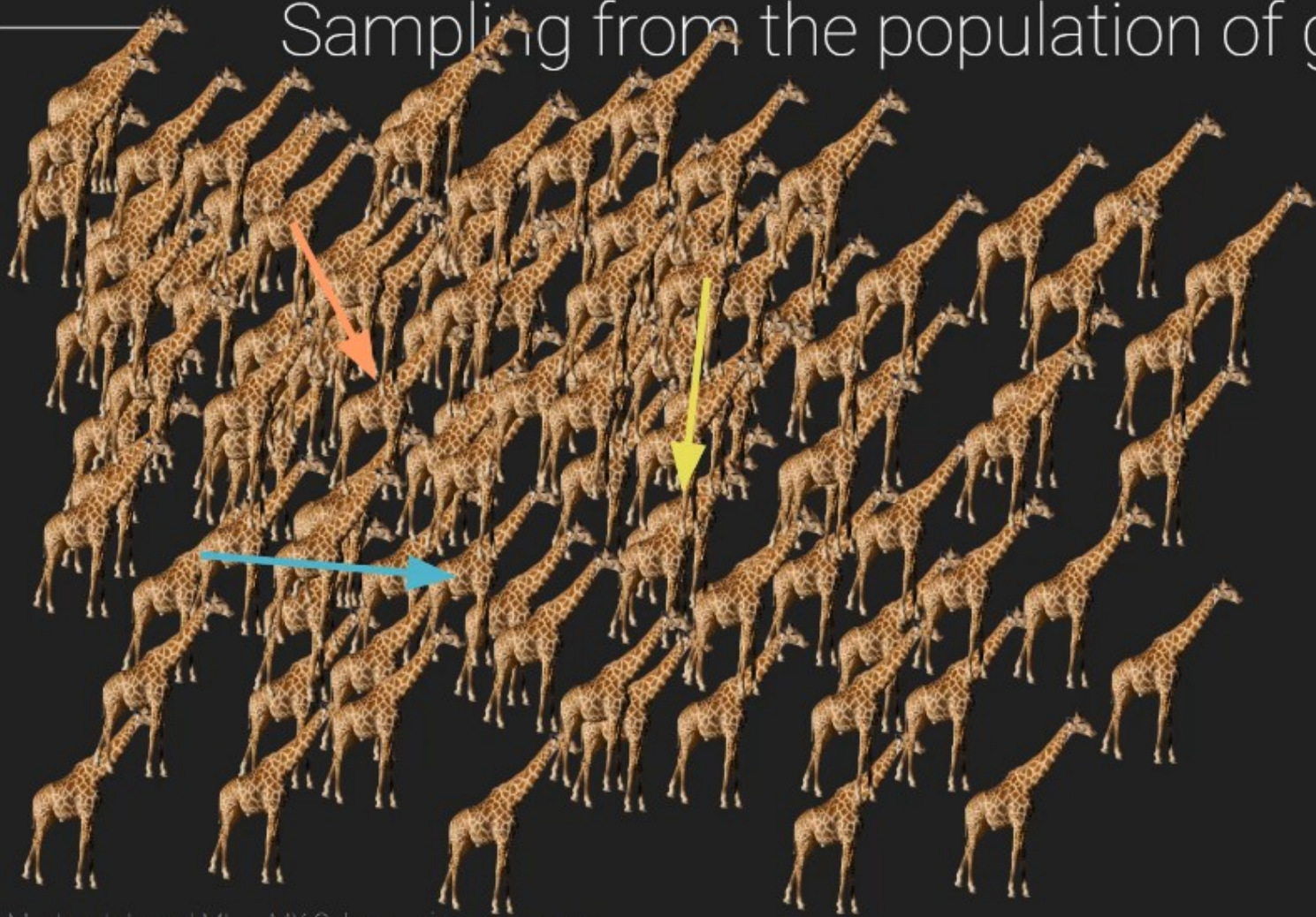


www.facebook.com › SanDiegoZoo › posts › a-group-of-giraffes-is-c... ▼

San Diego Zoo - A group of giraffes is called a "tower," a ...

Instead, we measure the heights of a sample of giraffes.

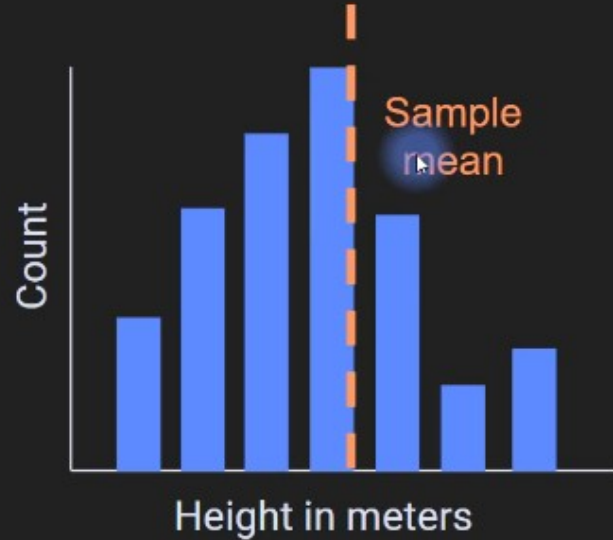
Sampling from the population of giraffes



g#	m
1	5.1
2	5.9
⋮	⋮
100	4.7

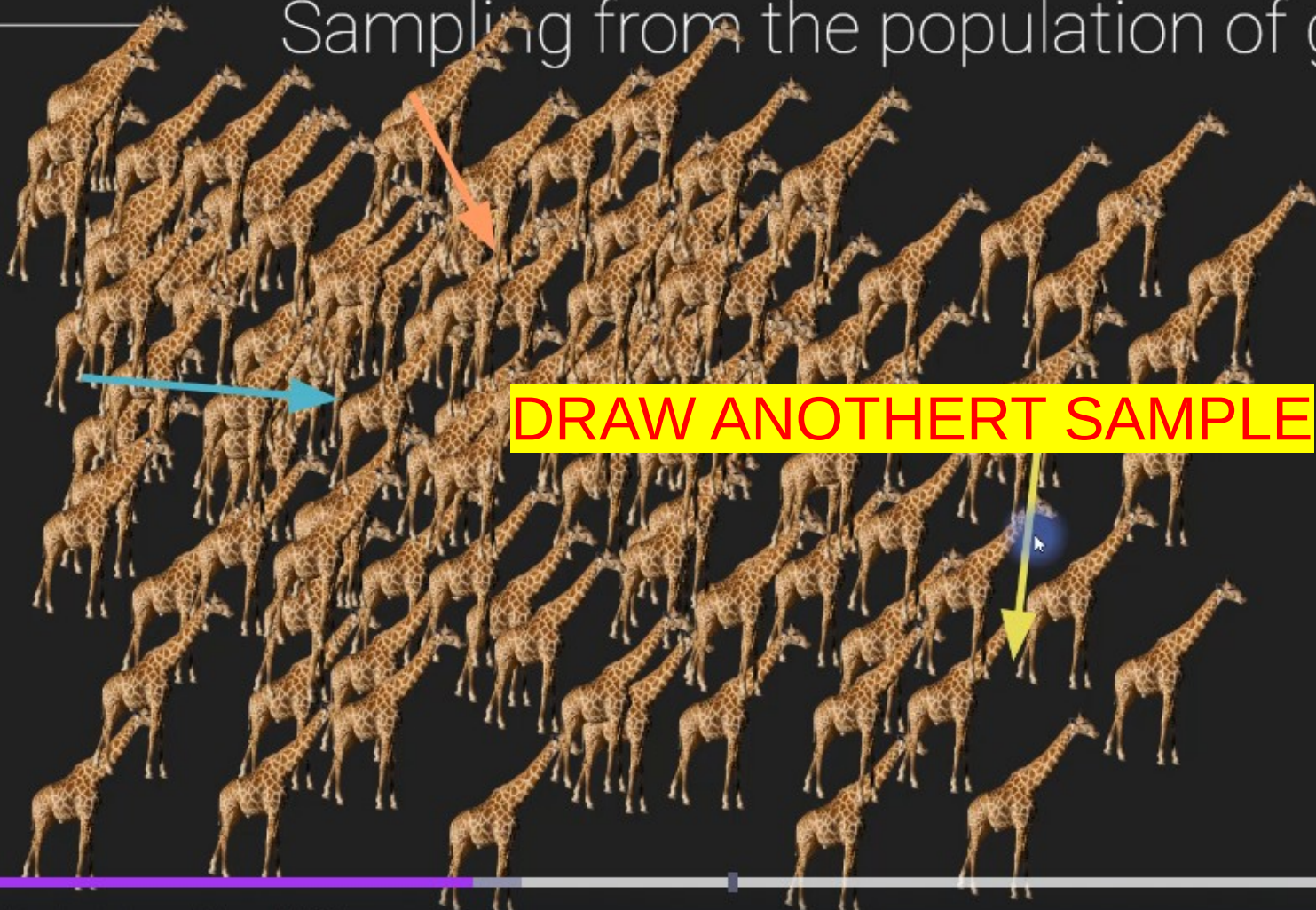
Sampling from the population of giraffes

IMPORTANT: This is a data distribution that produces **ONE** sample estimate (mean).



g#	m
1	5.1
2	5.9
⋮	⋮
100	4.7

Sampling from the population of giraffes

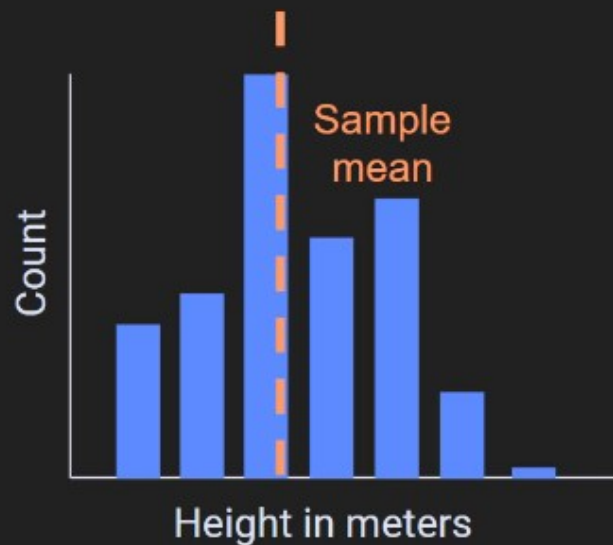


DRAW ANOTHER SAMPLE

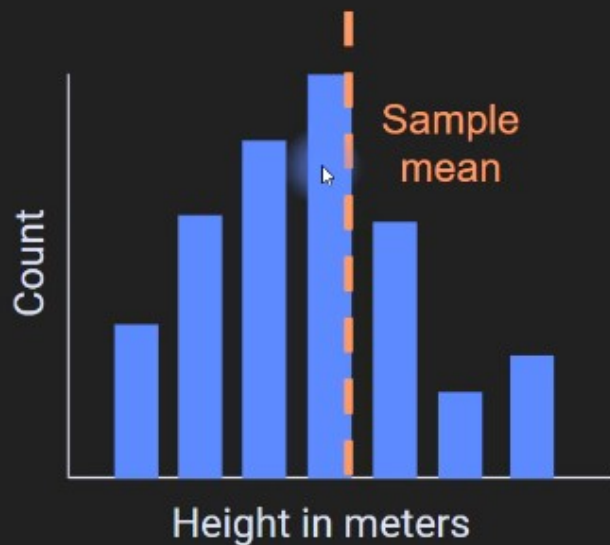
g#	m
1	6.2
2	6.0
⋮	⋮
100	5.6

Sampling from the population of giraffes

Data distribution and mean from sample 1



Data distribution and mean from sample 2

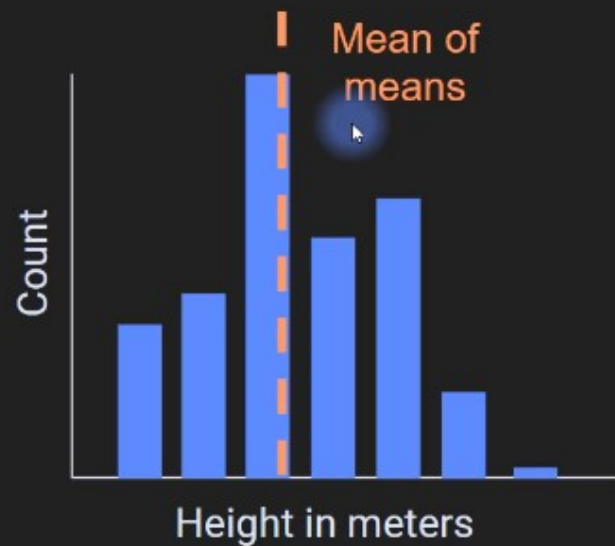


Data distribution and mean from sample N

...

Creating a sample distribution

Sample #	mean
1	5.1
2	5.8
⋮	⋮
50	6.7



Creating a sample distribution

A distribution of data from one sample has one mean.
Many means from many samples gives a sample distribution.

(Note: “mean” is just an example of a parameter estimate.)

Creating a sample distribution of differences

Scientific question: Are giraffes taller than slow lorises?



vs.



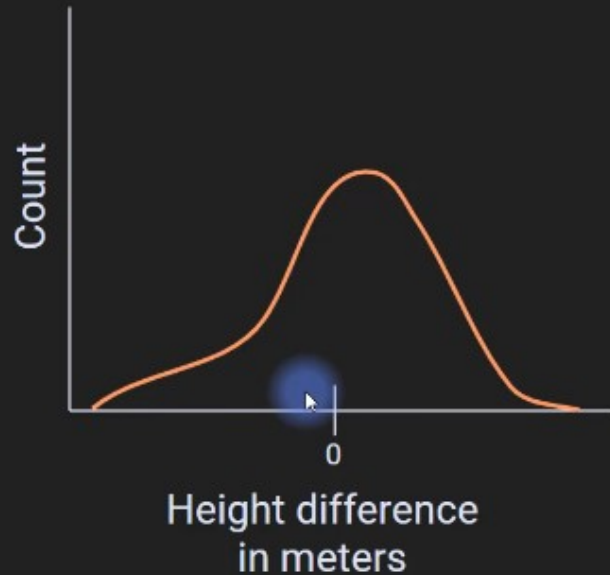
Creating a sample distribution of differences

Scientific question: Are giraffes taller than slow lorises?



Creating a sample distribution of differences

Scientific question: Are north African giraffes taller than south African giraffes?



Random, representative sampling

Scientific question: Is it warmer in the Netherlands than in Vietnam?

	NL	VN	
Successive days in July in Amsterdam	25	10	Successive days in January in Hanoi
	32	13	
	...	...	
	28	8	

Random, representative sampling

Scientific question: Is it warmer in the Netherlands than in Vietnam?

Appropriate answer based on sampling:
July in Amsterdam is warmer than
January in Hanoi.

Larger point: Sample estimates can be generalized only to the population that the sample represents.

Monte Carlo sampling

In this video, you will learn

- What “Monte Carlo sampling” means and how it relates to random sampling.

Monte Carlo methods

Monte Carlo methods: solve really hard problems by randomly sampling the solution space instead of doing *the real work*.

Often used in physics, statistics, deep learning, even in pure math.

“Monte Carlo sampling” is the same thing as randomly sampling from a population to estimate an unknown population parameter.

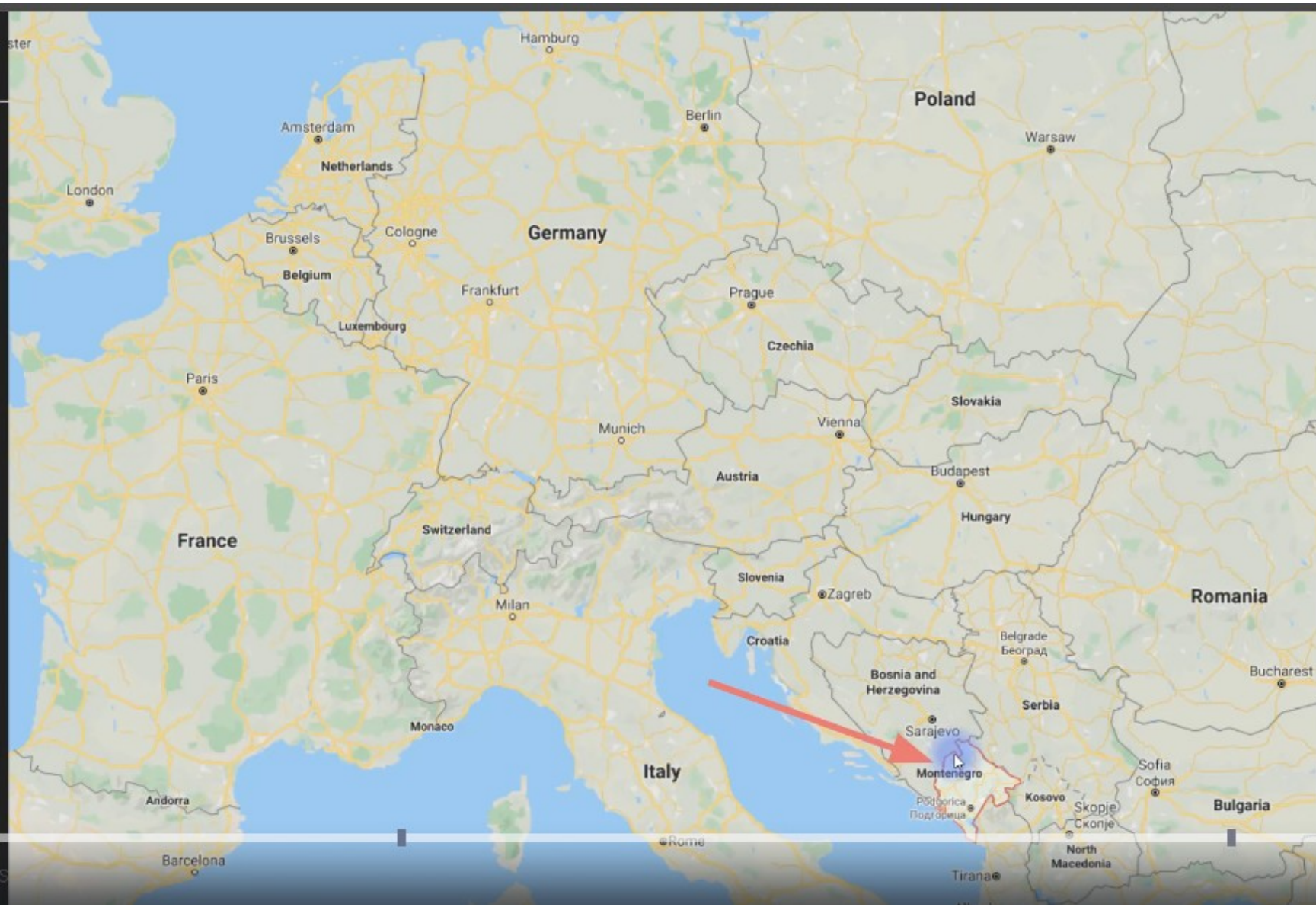
Sampling variability, noise, and other annoyances

In this video, you will learn

- The definition of sampling variability, where it comes from, and why it is important to understand.
- **Other sources of variability in measured data.**

Sampling variability: example

Scientific question:
How tall is the average Montenegrin?



Sampling variability: example

Scientific question:
How tall is the average Montenegrin?

Don't worry about the actual answer.

The question is: *How do we know the answer?*

